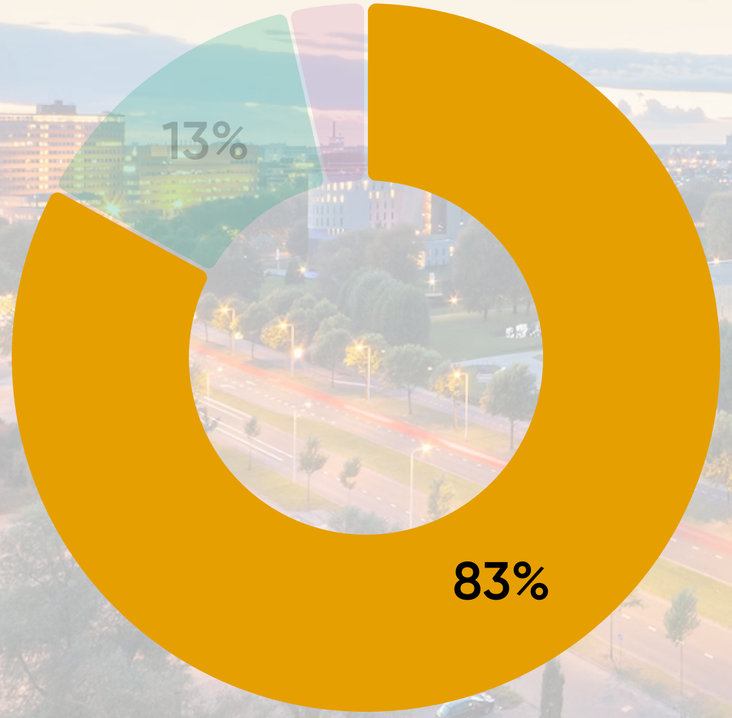
Multiple choice questions: Lectures 1&2



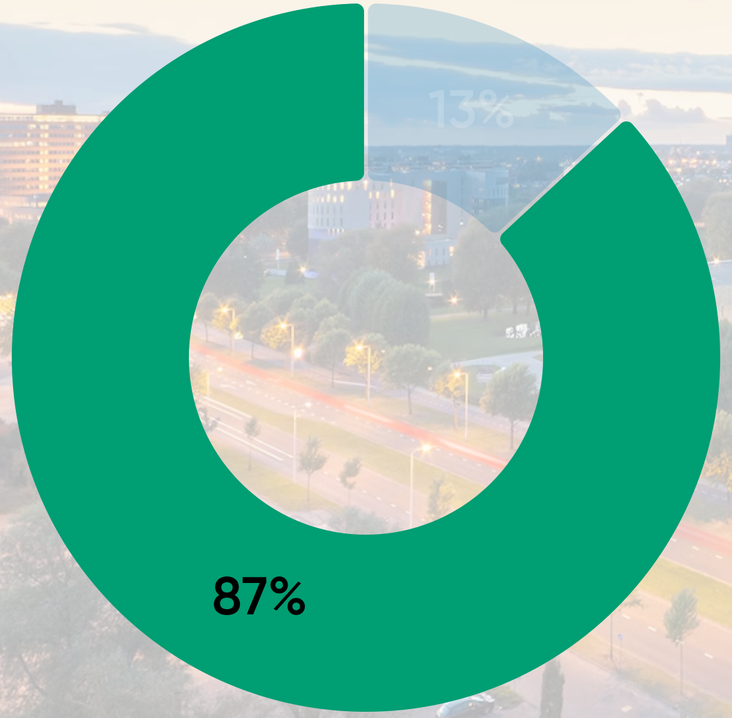
0

In autoregressive language model pretraining, how are the prediction targets obtained?



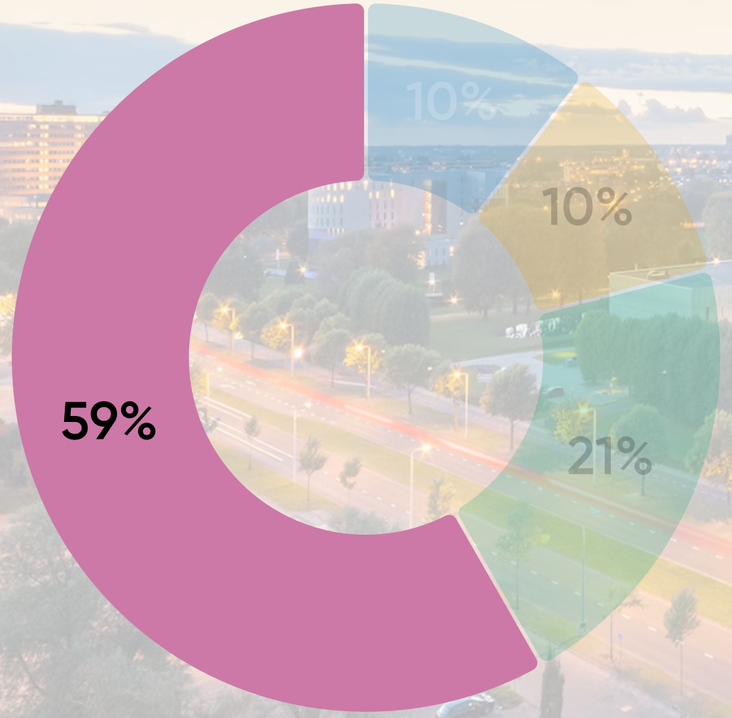
- 0% A separate annotator (expert) assigns a class to each sentence. ×
- 83% Each observed token serves as the target following its preceding context. ✓
- 13% The model's own generated continuation is used as the target sequence. ×
- 3% Randomly hidden tokens are predicted using the tokens on both sides. ×

In $h_i^{(0)} = e(x_i) + p_i$, what information does p_i supply?



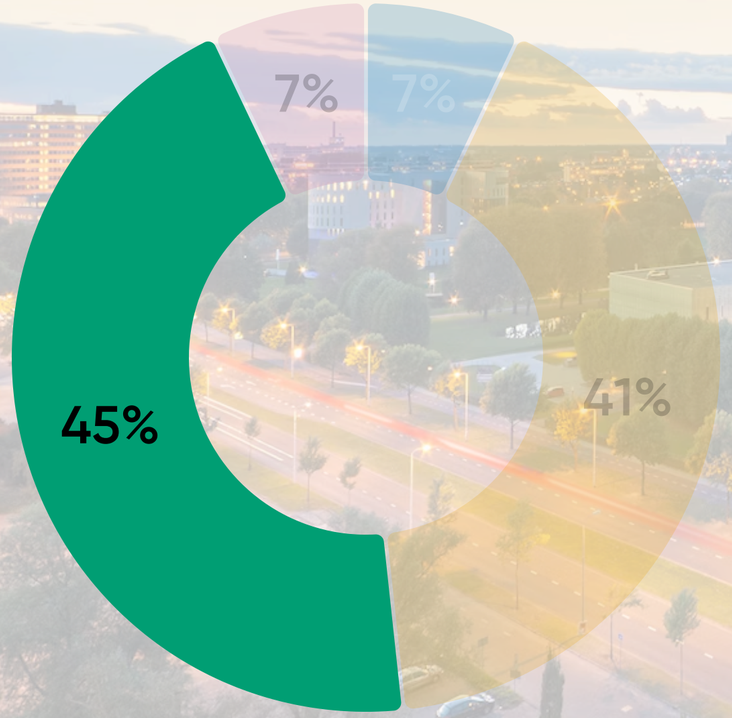
- 13% The probability of the token as the next output. ×
- 0% The number of times the token appears in the training set. ×
- 87% The position of the token in the sequence. ✓
- 0% The model's confidence that the token represents factual correct information. ×

What defines self-attention, as opposed to cross-attention?



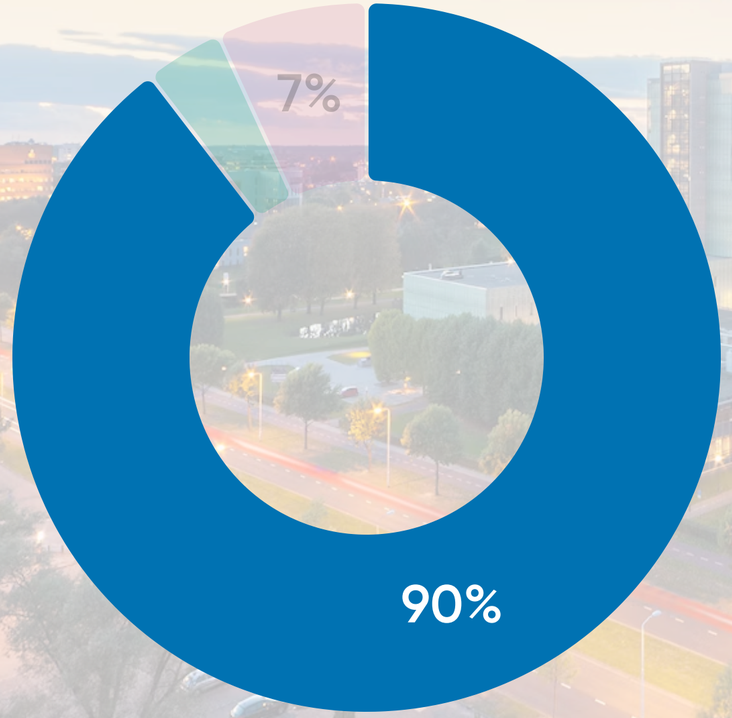
- 10% Every token can attend only to itself. ×
- 10% Attention is calculated without trainable parameters. ×
- 21% Queries, keys, and values use an identical learned projection matrix. ×
- 59% Queries, keys, and values are derived from the same input sequence. ✓

What distinguishes zero-shot from few-shot prompting?



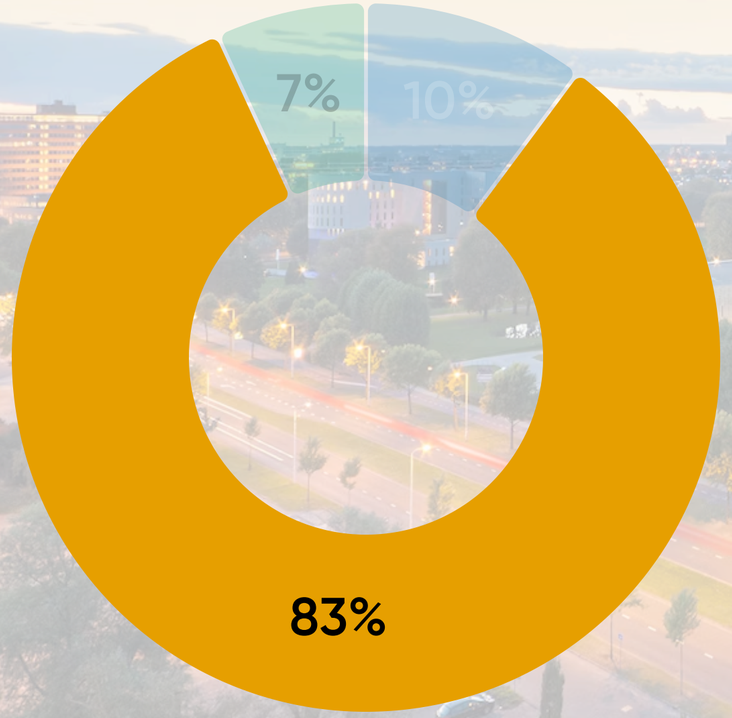
- 7% Zero-shot prompting freezes weights; few-shot prompting trains weights. ×
- 41% Zero-shot prompting uses no pretrained knowledge; few-shot prompting uses pretrained knowledge. ×
- 45% Zero-shot prompting supplies no task demonstrations; few-shot prompting supplies examples. ✓
- 7% Zero-shot prompting supplies no instruction; few-shot prompting supplies an instruction. ×

For $W \in \mathbb{R}^{d \times k}$ and $\Delta W = BA$ of rank at most r, which shapes are compatible?



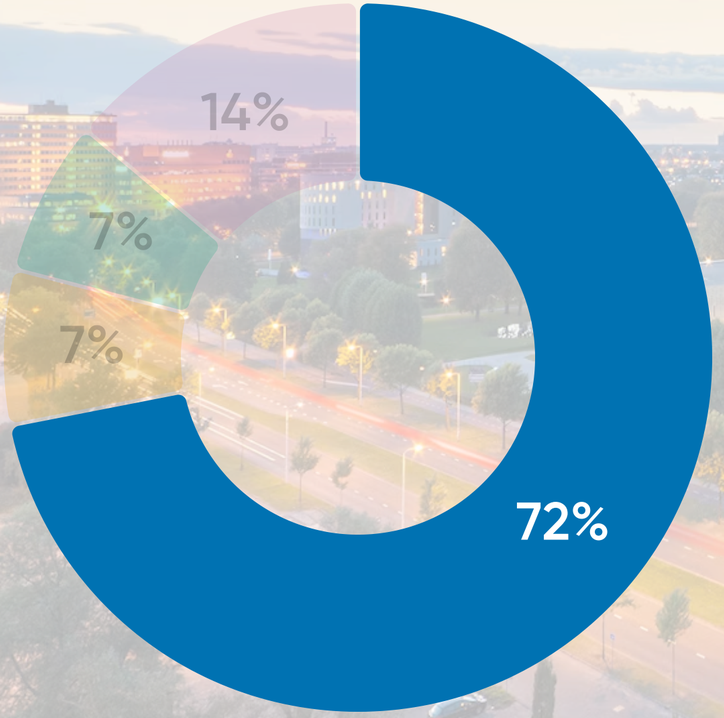
- 90% $B \in R^{d \times r}$ and $A \in R^{r \times k}$. ✓
- 0% $B \in R^{d \times k}$ and $A \in R^{r \times r}$. ×
- 3% $B \in R^{k \times r}$ and $A \in R^{r \times d}$. ×
- 7% $B \in R^{r \times d}$ and $A \in R^{k \times r}$. ×

Why are different augmented views of an image used in visual self-supervision?



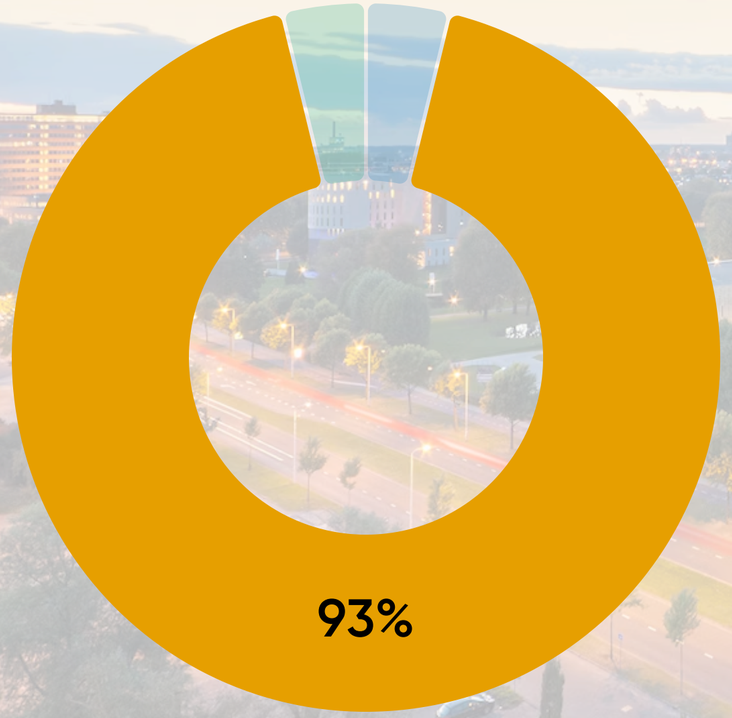
- 10% They guarantee preservation of every clinically important detail. ×
- 83% They provide related observations from which to learn consistent representations. ✓
- 7% They ensure all images have identical feature vectors. ×
- 0% They replace the need for an optimization objective. ×

What is the purpose of a contrastive image-text training objective?



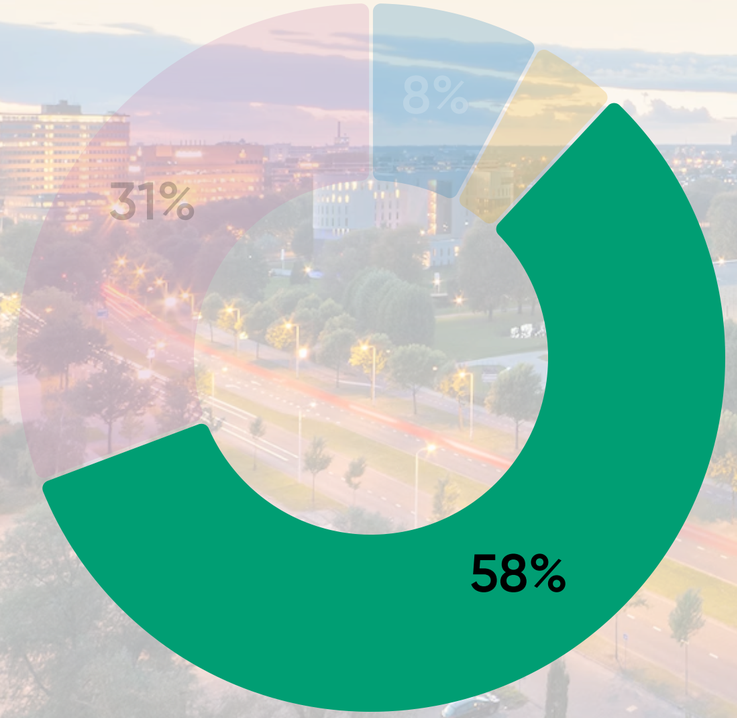
- 72% To give matching image–text pairs higher similarity than mismatched pairs. ✓
- 7% To predict successive words of a caption from its associated image. ×
- 7% To reconstruct image pixels from their associated text descriptions. ×
- 14% To assign images and text independently to a fixed set of diagnostic labels. ×

Which statement distinguishes image-to-report retrieval from report generation?



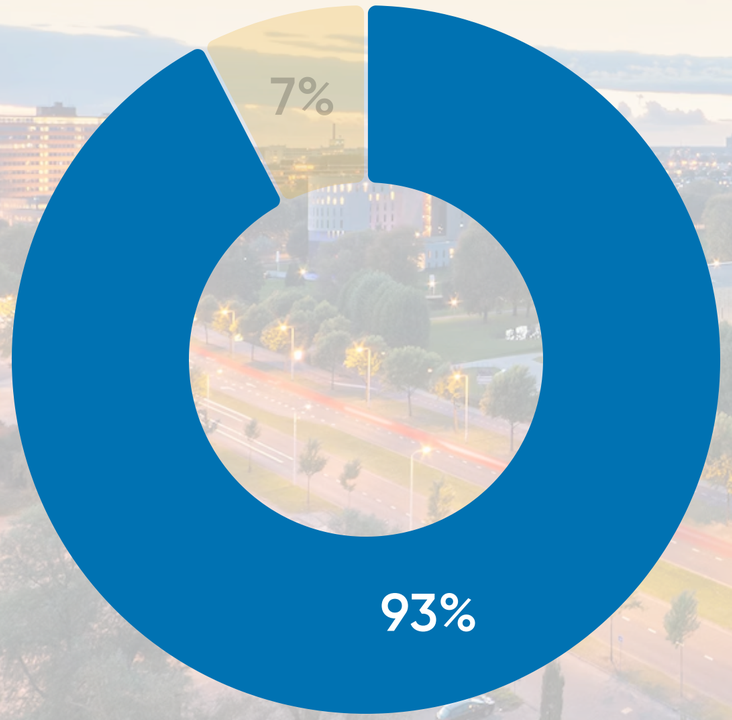
- 4% Retrieval predicts report tokens; generation ranks stored reports. ×
- 93% Retrieval selects an existing report; generation predicts a sequence of report tokens. ✓
- 4% Retrieval uses a text query only; generation uses an image query only. ×
- 0% Retrieval and generation both select stored reports but use different similarity metrics. ×

For fixed image features $z = f_\theta(x)$, a linear probe uses $p = \mathrm{softmax}(Wz + b)$. Which parameters are fitted?



- 8% θ, W, and b jointly. ×
- 4% No parameters are fitted, the class probabilities are known before training. ×
- 58% W and b, while θ is fixed. ✓
- 31% θ only, while W and b are fixed. ×

What is external validation of a biomedical AI model (as opposed to evaluation with an internal held-out test)?



- 93% Evaluation on independently collected data from outside the model-development source. ✓
- 7% Evaluation on a random subset withheld from the same development dataset. ×
- 0% Evaluation after selecting the best model on the original validation subset. ×
- 0% Evaluation on augmented copies of images used to train the model. ×